

## LM05 Natural Resources

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## 1. Introduction

Natural resources include:

- **Commodities:** Can be further classified into:
  - Hard: Commodities that are mined e.g., copper, gold, silver; and commodities that are extracted e.g., crude oil, natural gas.
  - Soft: Commodities that are grown over a period of time e.g., grains, livestock, and cash crops like coffee.
- **Farm land:**
  - Investments in land used for the cultivation of crops or livestock.
  - Income can be generated from the growth, harvest and sale of crops or livestock; or by leasing the land back to farmers.
- **Timberland:**
  - Investments in natural forests or managed tree plantations.
  - The return comes from the sale of trees, wood, and other timber products.

Up to about 20 years ago, investors looking for exposure to natural resources invested mainly via financial instruments (stocks and bonds). Instead of investing in the physical land and the products that come from it, investors focused on the companies that produced natural resources. Nowadays, however, due to the wide variety of direct investments available (ETFs, limited partnerships, REITS, swaps, and futures), investors typically participate in these assets directly.

This learning module covers:

- Features and investment characteristics of raw land, timberland, and farmland
- Features and investment characteristics of commodities
- Sources of risk, return, and diversification benefits of natural resource investments

## 2. Natural Resources Investment Features

### Timberland

Timberland provides an income stream through the sale of trees, wood, and other timber products. Timberland can be thought of as both a factory and a warehouse. The trees can be easily stored by simply not harvesting them. The trees can be harvested based on the price: more harvest when prices are up and delayed harvest when prices are down.

The three return drivers for timberland investments include: biological growth, change in prices of lumber (cut wood), and underlying land price change.

Additionally, since trees consume carbon as part of their life cycle, timberland considered a sustainable investment that mitigates climate-related risks.

**Farmland**

Farmland is perceived to provide a hedge against inflation. Two types of farm crops include raw crops that are planted and harvested, and permanent crops that grow on trees. Like timberland, farmland also provides an income component related to harvest quantities and agricultural commodity prices. However, it does not provide production flexibility, as farm products must be harvested when ripe.

Similar to timber land, the return drivers for farmland are: harvested quantities, commodity prices, and land price appreciation.

Farmland is also considered a sustainable investment that mitigates climate-related risks.

**Land Investments vs. Real Estate**

Farmland, timberland, and raw land are similar to real estate investments in that they are unique, illiquid assets with distinct geographic location and features. However, there are also some notable differences:

- Unlike real estate, there is no focus on the physical development of the land. Rather than development, the quality of the soil and climate features matter more.
- The location of the land is important. The closer it is to transportation hubs and markets, the higher the price. This is because transportation expenses can be a significant component of the price of the products paid by the customers of timberland and farmland.

To invest in raw land, timberland, and farmland, investors need specialized knowledge and understanding of the specifics of the natural resources. For example, to invest directly in timberland one needs forest expertise to manage a forest over its life cycle. Many institutional investors do not have this expertise, and rely on timberland investment management organizations (TIMOs)- TIMOs are entities that use their forest investment expertise to analyze and acquire suitable timberland holdings on behalf of institutional investors.

Exhibit 1 from the curriculum outlines the features of land investments.

|                          | <b>Raw land</b>                       | <b>Farmland</b>                                                                      | <b>Timberland</b>                                                                     |
|--------------------------|---------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Return drivers           | Price of land                         | Harvest quantities<br>Commodity prices<br>Price of land                              | Biological growth<br>Harvest quantities<br>Lumber prices<br>Price of land             |
| Source of direct revenue | Price appreciation<br>Lease revenue   | Sale of crops and other agricultural products<br>Price appreciation<br>Lease revenue | Sale of trees, wood, and other timber products<br>Price appreciation<br>Lease revenue |
| Value                    | Physical location                     | Physical location<br>Growth cycle<br>Soil quality                                    | Physical location<br>Quality of timber<br>Phase in timber production                  |
| Main risks               | Best alternative use                  | Weather factors and climate change<br>Biological factors, diseases                   |                                                                                       |
| Owners                   | Mostly institutional, some individual | Mostly individuals, some institutional                                               | Mostly institutional, some individual                                                 |
| Ownership structure      | Direct ownership, partnership         | Direct ownership, partnership, REIT                                                  | Direct ownership, partnership, REIT, TIMO                                             |

### Features and Forms of Farmland and Timberland Investment

Timberland tracts are typically thousands (or more) of acres in size, whereas farmland is frequently owned in smaller tracts of tens or hundreds of acres. As a result, farmland is more suited to family ownership, whereas timberland is more commonly owned by institutions.

An advantage of institutional ownership of physical farmland, as compared to exposure to crops through futures contracts is: Futures contracts are only available on a few common crops (such as wheat and corn). Ownership of physical farmland allows for the cultivation of crops that are not traded on futures exchanges, resulting in a broader universe of agricultural product price exposures.

However, the disadvantage is: there is limited price transparency or information to guide investment decisions; assistance from sector specialists is required. Direct investments are illiquid.

Farmland is highly sensitive to weather changes. Bad weather conditions can drastically reduce harvest yields.

### 3. Commodity Investment Forms

Commodities are physical products that can be standardized on quality, location, and delivery for investment purposes.

Generally, commodity investments take place through derivative instruments, because of the high storage and transportation costs incurred when holding commodities physically. The underlying asset of a commodity derivative may be a single commodity or an index of commodities. The return on commodity investment is based mainly on price changes rather than an income stream such as dividends.

In order to be transparent, investable, and replicable, commodity indexes typically use the price of the futures contracts rather than the prices of the underlying physical commodities.

Commodity sectors include:

- Energy - oil, natural gas, coal, electricity etc.
- Base metals - copper, aluminum, zinc etc.
- Precious metals - gold, silver, platinum etc.
- Agriculture - grains, livestock, coffee etc.
- Others - carbon credits, freight, forest products etc.

#### How are commodity futures contracts priced?

- The price of a futures contract can be calculated using the following formula:  

$$\text{Future price} \approx \text{Spot price} (1 + r) + \text{Storage costs} - \text{Convenience yield}$$
 where: convenience yield is the value associated with holding the physical asset;  
 $r$  is the short-term risk-free interest rate
- Future prices may be higher or lower than spot prices, based on convenience yield.
- For no arbitrage to occur,  $\text{Future price} \approx \text{Spot price} (1+r)$ . But commodities incur storage costs. So, they must be added to the future price and we get  $\text{Future price} \approx \text{Spot price} (1 + r) + \text{storage costs}$ . Storage and interest costs are collectively known as “cost of carry”.
- Why subtract the convenience yield? Because the buyer does not possess the commodity as of now, until the end of the contract. Since he has given up this convenience, it must be subtracted from the future price. That’s how we arrive at  $\text{Future price} \approx \text{Spot price} (1 + r) + \text{storage costs} - \text{convenience yield}$
- Futures price may be higher or lower than the spot price based on the convenience yield.

Contango:  $\text{Future price} > \text{Spot price}$  Markets tend to be in contango when there is little or no convenience yield.

Backwardation:  $\text{Future price} < \text{Spot price}$  Markets tend to be in backwardation when the convenience yield is high.

Impact of inventory levels: When the inventory levels of a specific commodity are low, market participants would prefer to own the physical commodity rather than a derivative contract. This incentive raises spot prices relative to forward prices, resulting in a backwardation.

### Forms of Commodity Investments

Commodity investments are typically made through derivatives as the storage and transportation costs for holding physical commodities are significant. Commodity derivative contracts may trade on exchanges or over the counter. The popular derivatives include futures, forwards, options, and swaps.

Commodity exposure can also be achieved through:

- Exchange traded products (either funds or notes): Suitable for investors seeking simplified trading through a standard brokerage account.
- Managed futures (also known as CTAs): Commodity trading advisers create trading strategies based on commodity derivative contracts that are aimed at predicting upcoming bull or bear trends.
- Funds that specialize in specific commodity sectors: e.g., private energy partnerships are similar to PE funds and can be used to gain exposure to the energy sector.

## 4. Natural Resource Investment Risk, Return, and Diversification

### Commodities

Commodities offer potential for high returns, portfolio diversification, and inflation protection.

Commodity spot prices are a function of supply and demand, the costs of production and storage, value to users, and global economic conditions.

- Supplies of commodities depend on production and inventory levels.
- Demand of commodities depends on the consumption needs of end users.
- Demand may be high while supply may be low during economic growth; conversely, demand may be low and supply high during times of economic slowdown.
- If demand changes very quickly during any period, resulting in supply-demand mismatch, it may lead to price volatility.

Typically, commodity investments are made through derivative contracts, which are highly leveraged financial instruments. As a result, the observed returns are extremely volatile.

Commodities are attractive to investors not only for the potential profits but also because:

- Commodities exhibit high correlation with inflation over the last 30 years, suggesting that commodities are an effective inflation hedge. Some commodity prices are a component of inflation calculations e.g., food and energy.

- They provide effective portfolio diversification. Historically, the correlation between commodities and traditional investments has been low.

### **Farmland and Timberland**

Timberland and farmland investments have similar risks as other real estate investments in raw land. However, weather is major risk factor for these investments. Droughts and flooding can dramatically decrease yields.

Another important risk factor is the international competitive landscape. Unlike other real estate that is mainly impacted by local factors, timberland and farmland produce commodities that are globally traded; therefore, they are impacted by global factors.

Farmland and timberland investments are traded infrequently and in private markets. As a result, despite the fact that both asset classes face significant risks, such as weather-related threats, they are likely to appear less volatile than commodities and other publicly traded risky assets (such as stocks).

ESG investors looking for responsible and sustainable investing can include timberland and farmland in their portfolios. These investments can help mitigate climate change.

Timberland and farmland have also exhibited low correlation with traditional investments. Thus, they can provide effective diversification benefits.

## Summary

**LO: Explain features of raw land, timberland, and farmland and their investment characteristics.**

|                          | <b>Raw land</b>                       | <b>Farmland</b>                                                                      | <b>Timberland</b>                                                                     |
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**LO: Describe features of commodities and their investment characteristics.**

Generally, commodity investments take place through derivative instruments, because of the high storage and transportation costs incurred when holding commodities physically.

The return on commodity investment is based mainly on price changes rather than an income stream such as dividends.

**LO: Analyze sources of risk, return, and diversification among natural resource**



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